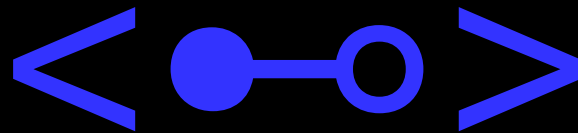


Dezvoltarea aplicațiilor Web



date structurate în cadrul documentelor HTML
RDFa – declarații de triple RDF în paginile Web

Prof. Sabin Corneliu Buraga
profs.info.uaic.ro/sabin.buraga/

„Dacă poți acorda ajutor, ajută-i pe ceilalți.
Dacă nu poți, măcar nu-i deranja.”

Dalai Lama



Folosind modelul RDF (subiect, predicat, obiect),
n-am putea include triplele RDF în paginile Web?

Specificarea construcțiilor RDF direct în (X)HTML **RDFa**

reprezintă un alt format de serializare a modelului RDF

$$\mathbf{RDF + HTML = RDFa}$$

RDFa

recomandări ale Consorțiului Web

RDFa 1.0 (2008)

RDFa 1.1 (2012 – ediția 1; 2013 – ediția 2; 2015 – ediția 3)

www.w3.org/TR/rdfa-core/

RDFA

„punte” între Web-ul social și Web-ul datelor

www.w3.org/TR/rdfa-primer/

RDFA

utilizarea unor attribute HTML specifice menite a modela
datele disponibile într-o pagină Web
în vederea procesării acestora de către un program

RDFA

minimal, se pot utiliza attributele
precizate de recomandarea **RDFA Lite 1.1** (W3C, 2015)

www.w3.org/TR/rdfa-lite/

RDFa

vocab – desemnează URL-ul vocabularului de date
(modelului conceptual) adoptat

RDFa

vocab – desemnează URL-ul vocabularului de date
(modelului conceptual) adoptat

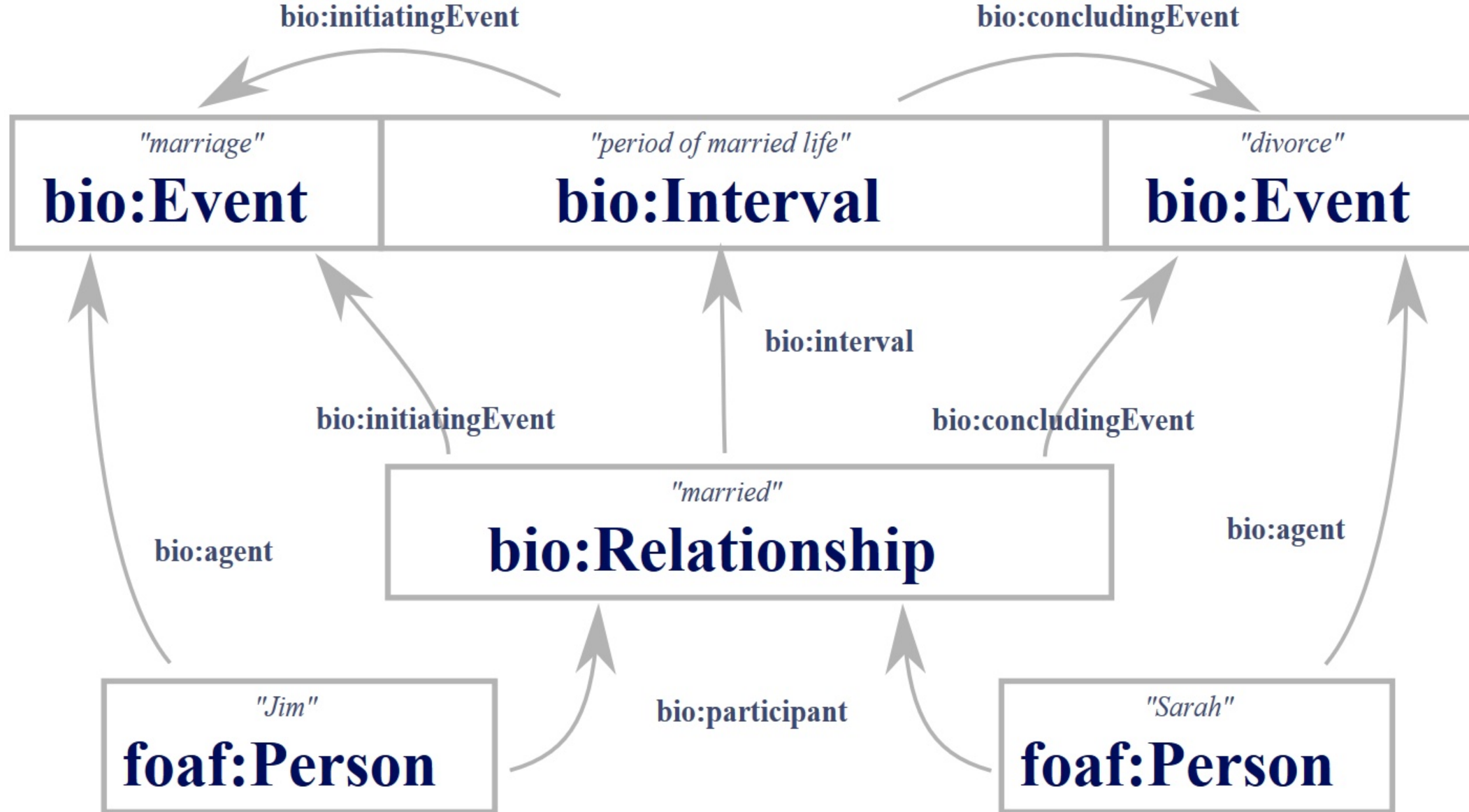
exemplificări:

concepte definite de schema.org

meta-date via DCMi (*Dublin Core Metadata Initiative*)

diverse vocabulare utile – vocab.org

cunoștințe vizând știri (NewsML, rNews) – dev.iptc.org



Biography Vocabulary Core Classes

<http://purl.org/vocab/bio/0.1/>
by Ian Davis, June 2011

studiu de caz:

BIO – vocabular privind biografia unei persoane

vocab.org/bio/

RDFa

typeof – precizează un tip de date (clasă de „lucruri”)

exemple:

folosind modelul conceptual specificat de schema.org

Country Event Image Organization Person Product Project

RDFa

property – specifică o proprietate asociată tipului de date

exemplul #1:

pentru tipul **Event** sunt definite proprietățile
duration, location, startDate, endDate, performers,...
(conform <http://schema.org/Event>)

RDFa

property – specifică o proprietate asociată tipului de date

exemplul #2:

pentru conceptul **Project** sunt specificate proprietățile **repository**, **wiki**, **download-page**, **developer**, **documenter** etc.
(conform vocabularului DOAP – *Description Of A Project*)

RDFa

resource – indică un identificator de resursă (subiectul)
despre care se „vorbește” în termeni de meta-date

RDFa

prefix – desemnează prefixul unui vocabular specific,
facilitând referirea acestuia

RDFa

prefix – desemnează prefixul unui vocabular specific, facilitând referirea acestuia

exemplificări tipice:

rdf – aserțiuni RDF

rdfa – construcții RDFa

xsd – tipuri de date specificate de XML Schema

...

Prefix	URI	Description
as	https://www.w3.org/ns/activitystreams#	Activity Vocabulary
dcat	http://www.w3.org/ns/dcat#	Data Catalog Vocabulary
ldp	http://www.w3.org/ns/ldp#	Linked Data Platform Voc.
ma	http://www.w3.org/ns/ma-ont#	Media Resources
org	http://www.w3.org/ns/org#	Organizations
owl	http://www.w3.org/2002/07/owl#	OWL
prov	http://www.w3.org/ns/prov#	Provenance Vocabulary
qb	http://purl.org/linked-data/cube#	Data Cubes
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#	RDF
rdfa	http://www.w3.org/ns/rdfa#	RDFA Vocabulary
rdfs	http://www.w3.org/2000/01/rdf-schema#	RDF Schema
skos	http://www.w3.org/2004/02/skos/core#	SKOS Core
ssn	http://www.w3.org/ns/ssn/	Semantic Sensor Network
time	http://www.w3.org/2006/time#	Time Ontology
xsd	http://www.w3.org/2001/XMLSchema#	XML Schema Datatypes

lista prefixelor predefinite:
www.w3.org/2011/rdfa-context/rdfa-1.1

Exemplu recurgând la modelul conceptual schema.org adaptare după Manu Sporny

```
<p vocab="http://schema.org" typeof="Person"
  resource="#tux">
  Salut, eu sunt
  <span property="name">Tuxy Pinguinnesscool</span>
  și vă invit să vizitați
  <a property="url" href="http://tux.info/"
    title="Situl Web al lui Tuxy">situl meu</a>.
</p>
```

adnotare explicită direct în HTML:
resursa identificată cu **tux** este o persoană având atașate
un nume și o adresă Web a sitului personal

Exemplu recurgând la modelul conceptual schema.org adaptare după Manu Sporny

```
<p vocab="http://schema.org" typeof="Person"
  resource="#tux">
  Salut, eu sunt
  <span property="name">Tuxy Pinguinnesscool</span>
  și vă invit să vizitați
  <a property="url" href="http://tux.info/"
    title="Situl Web al lui Tuxy">situl meu</a>.
</p>
```

```
@prefix s: <http://schema.org/> .
:tux      rdf:type      s:Person ;
          s:name        "Tuxy Pinguinnesscool" ;
          s:url          <http://tux.info/> .
```

asertiunile RDF
echivalente

Exemplu: modelarea grafului social pe baza modelului schema.org și vocabularului **Relationship**

```
<div vocab="http://schema.org/">
  <div resource="#busaco" typeof="Person">
    <a property="url" href="https://profs.info.uaic.ro/~busaco/">
      <span property="name">
        <span property="firstName">Sabin</span>
        <span property="lastName">Buraga</span>
      </span></a>
    <a property="http://purl.org/vocab/relationship/collaboratesWith"
      href="#acip">colaborează cu Ciprian</a>.</div>
  <p resource="#acip" typeof="Person">
    <span property="name"><span property="title">MSc.</span>
    Ciprian Amariei</span> predă la
    <span resource="#web" typeof="http://schema.org/Course">
      <span property="courseCode">Web</span></span>.&br/></p>
</div>
```

```
<div vocab="http://schema.org/">
  <div resource="#busaco" typeof="Person">
    <a property="url" href="https://profs.info.uaic.ro/~busaco/">
      <span property="name">
        <span property="firstName">Sabin</span>
        <span property="lastName">Buraga</span>
      </span></a>
    <a
      property="http://purl.org/vocab/relationship/colaboratesWith"
      href="#acip">colaborează cu Ciprian</a>.</div>
    <p resource="#acip" typeof="Person">
      <span property="name"><span property="title">MSc.</span>
        Ciprian Amariei</span> predă la
      <span resource="#web" typeof="http://schema.org/Course">
        <span property="courseCode">Web</span></span>.</p>
  </div>
```

Sabin Buraga colaborează cu Ciprian

MSc. Ciprian Amariei predă la Web.

redarea
conținutului
marcat

date adnotate
semantic via RDFa

rdfa.info/play/

un instrument Web de editare și vizualizare
a construcțiilor RDFa

@prefix rdfs: <<http://www.w3.org/ns/rdfs/>> .

@prefix schema: <<http://schema.org/>> .

@prefix rdf: <<http://www.w3.org/1999/02/22-rdf-syntax-ns#>> .

<<http://rdfs.info/play/>> rdfs:usesVocabulary schema: .

<<http://rdfs.info/play/#busaco>>

rdf:type schema:Person;

schema:url <<https://profs.info.uaic.ro/~busaco/>>;

schema:name "Sabin Buraga";

schema:firstName "Sabin"; schema:lastName "Buraga";

<<http://purl.org/vocab/relationship/colloboratesWith>>

<<http://rdfs.info/play/#acip>> .

<<http://rdfs.info/play/#acip>>

rdf:type schema:Person;

schema:name "MSc. Ciprian Amariei";

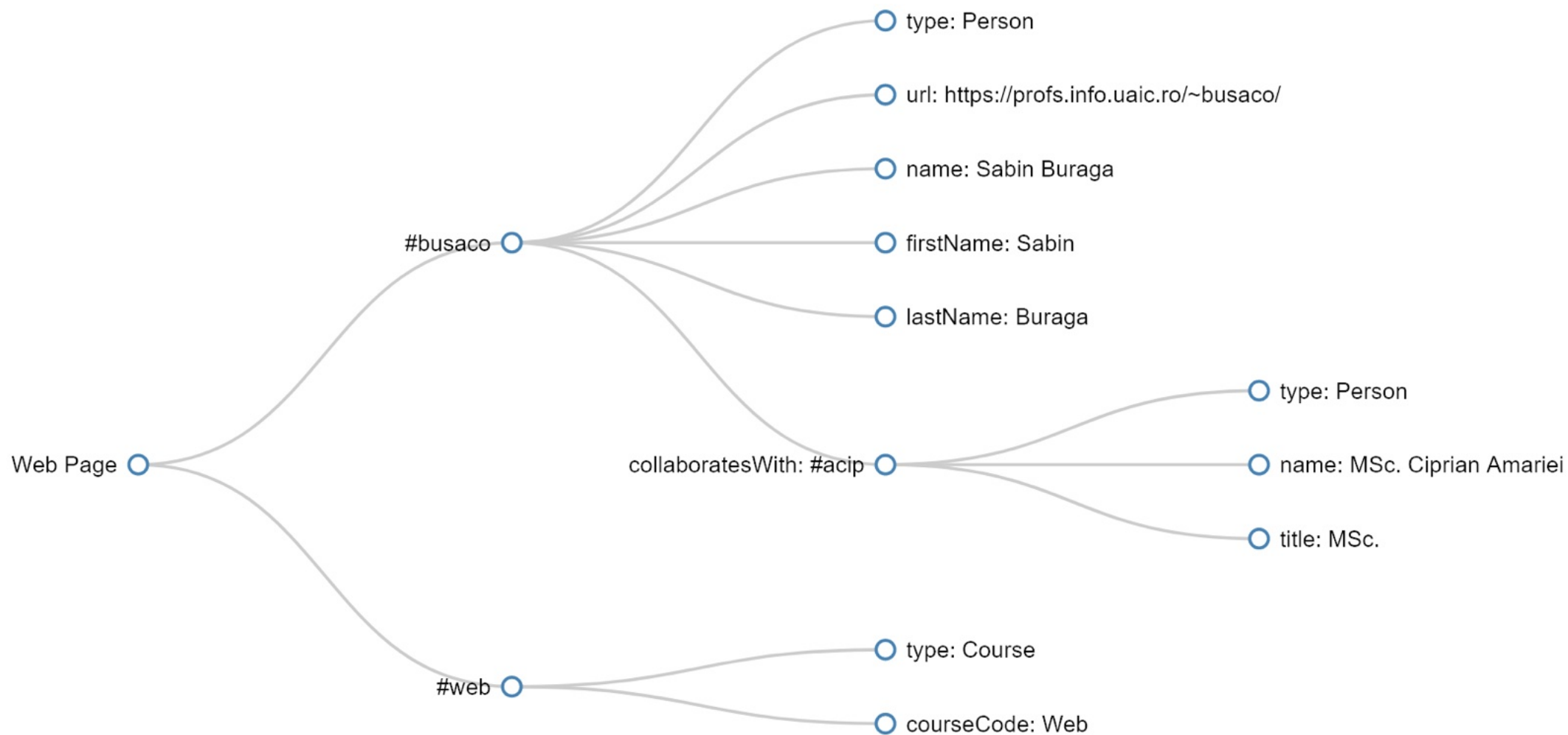
schema:title "MSc." .

<<http://rdfs.info/play/#web>>

rdf:type schema:Course;

schema:courseCode "Web" .

triplele RDF generate



vizualizarea grafului RDF corespunzător

Similar microdatelor HTML,
RDFa transformă o pagină Web într-un API

Similar microdatelor HTML,
RDFa transformă o pagină Web într-un API

- ▶ acces la date structurate – „purtând” un înțeles –,
nu doar la conținut textual
(*things, not strings*)

rdfa.info

RDFA transformă o pagină Web într-un API

diverse biblioteci/module de prelucrare:

clj-rdfa (Closure), **EasyRDF** (PHP),
N3.js (JavaScript), **node-rdf** (JavaScript),
PyRDFa (Python), **rdflib.js** (JavaScript)

rdfa.info/tools

rdfa.info/dev

vezi și **www.w3.org/community/rdfjs/wiki/Comparison_of_RDFJS_libraries**



Diverse studii de caz privitoare la RDFa?

RDFa 1.1 Distiller and Parser

Warning: This version implements [RDFa 1.1 Core](#), including the handling of the [Role Attribute](#). The distiller can also run in XHTML+RDFa 1.0 (XHTML content uses the RDFa 1.0 DTD and/or sets the `version` attribute). The [package available for download](#), although it may be slightly out of date, is still running this service.

Distill by URI

Distill by File Upload

Distill by Direct Text Input

Distill by URI

URI:

<https://pubmed.ncbi.nlm.nih.gov/32389849/>

Output Format:

Turtle

Returned content:

Only core triples

Expand vocabularies:

Yes

Generate warnings for non
RDFa 1.1 Lite usage:

No

► [More \(non-standard\) options](#)

Go!

URL-ul resursei
conținând (meta-)date
exprimate prin RDFa

If you intend to use this service regularly on large scale, [consider downloading the package](#) and use it locally. Storing a (conceptually)

extragerea cu instrumentul *RDFa 1.1 Distiller and Parser*
a triplelor RDF dintr-un document HTML ce include RDFa
github.com/RDFLib/pyRdfa3
aici, un articol științific oferit de *PubMed*

@prefix og: <<http://ogp.me/ns#>> .

subject

<<https://pubmed.ncbi.nlm.nih.gov/32389849/>>

og:description "Based on these early observations, as pediatricians..."@en ;

og:image "https://cdn.ncbi.nlm.nih.gov/.../pubmed-meta-image.png" ;

og:site_name "PubMed"@en ;

og:title "From SARS to COVID-19: What we have learned about
children infected with COVID-19 - PubMed"@en ;

og:type "website"@en ;

og:url "https://pubmed.ncbi.nlm.nih.gov/32389849/"@en .

object

predicat

pe baza construcțiilor RDFa specificate în codul HTML,
sunt extrase date RDF disponibile în format Turtle
în acest caz, se recurge la vocabularul **OGP** (*Open Graph
Protocol*) creat și utilizat de Facebook (Meta) – ogp.me

@prefix rdf: <<http://www.w3.org/1999/02/22-rdf-syntax-ns#>>.

@prefix schema: <<http://schema.org/>>.

detalii în cursul
despre RDF

<_:gs0>

rdf:type <<http://schema.org/WebPage>>;
schema:name "Web Application Development";
schema:description "syllabus (aka the WADe film)";
schema:license <<https://creativecommons.org/licenses/by-nc-sa/4.0/>>;
schema:author [a schema:Person ;
 schema:name "Sabin-Corneliu Buraga"@en ;
 schema:url <<https://profs.info.uaic.ro/sabin.buraga/>>] .

"#wade-lecture5"

rdf:type <<http://schema.org/CreativeWork>>;
schema:url <<https://profs.info.uaic.ro/~busaco/teach/.../web05ModelulRDF.pdf>>;
schema:name "Data Modeling in RDF (Resource Description Framework)";
schema:keywords "Web, model, RDF, data, vocabulary, ..., JSON-LD".

extragerea de microdate HTML5 și RDFa, cu convertirea în
triple RDF via **RDF Converter** – issemantic.net/rdf-converter
alternativă: converter.zazuko.com

```

{ "@context": { "schema": "http://schema.org/" },
  "@graph": [ {
    "@id": "_:N21752188d2184128ba182402e605ec5c",
    "@type": "schema:WebPage",
    "schema:author": { "@id": "_:N48eb473b6579441c817f39bd21fd8990" },
    "schema:description": "syllabus (aka the WADe film)",
    "schema:license": { "@id": "https://creativecommons.org/licenses/by-nc-sa/4.0/" },
    "schema:name": "Web Application Development"
  },
  { "@id": "_:N48eb473b6579441c817f39bd21fd8990",
    "@type": "schema:Person",
    "schema:name": "Sabin-Corneliu Buraga",
    "schema:url": { "@id": "https://profs.info.uaic.ro/sabin.buraga/" }
  },
  { "@id": "_:N35ffb1753f2247329028d3358c0ade52",
    "@type": "schema:CreativeWork",
    "schema:keywords": "Web, model, RDF, data, annotation, vocabulary, ..., JSON-LD",
    "schema:name": "Data Modeling in RDF (Resource Description Framework)",
    "schema:url": { "@id": "https://profs.info.uaic.ro/~busaco/.../web05ModelulRDF.pdf" }
  }
]
}

```

(meta-)date structurate serializate ca JSON-LD

← → ↻ www.drugs.com/coracidin.html

JSON-LD RDFa POSH

Drug Know me

Home > Coricid

Coricid

Fact Collection #1

Subject <https://www.drugs.com/coracidin.html>

Predicates

- [fb:app_id](#) "1470988726468809"@en
- [article:author](#) " <https://www.facebook.com/Drugscom> "@en
- [article:publisher](#) " <https://www.facebook.com/Drugscom> "@en
- [ogp:title](#) "Coricidin: Uses, Dosage & Side Effects - Drugs.com"@en
- [ogp:image](#) 

meta-date (proprietăți)
exprimate via **OGP**
title type description image url site_name

Uses | W

What is

- [ogp:site_name](#) "Drugs.com"@en
- [ogp:locale](#) "en_US"@en
- [ogp:type](#) "article"@en
- [ogp:description](#) "Coricidin (acetaminophen and chlorpheniramine) treats runny nose, sneezing, watery eyes, and pain or fever caused by allergies. Includes Coricidin side effects, interactions and indications."@en
- [ogp:url](#) " <https://www.drugs.com/coracidin.html> "@en

extensia de *browser* **OpenLink Structured Data Sniffer**
– osds.openlinksw.com – poate extrage și construcții RDFa

[ogp:image](#)

meta-date ale unui video OGP și alte vocabulare utilizate de YouTube

<http://ogp.me/ns#image:width>
<http://ogp.me/ns#image:height>
<ogp:description>

al:ios:app_store_id

al:ios:app_name

al:ios:url

al:android:url

al:web:url

[ogp:type](#)
<http://ogp.me/ns#video:url>
http://ogp.me/ns#video:secure_url
<http://ogp.me/ns#video:type>
<http://ogp.me/ns#video:width>
<http://ogp.me/ns#video:height>

al:android:app_name

al:android:package

<http://ogp.me/ns#video:tag>
<http://ogp.me/ns#video:tag>

<https://i.ytimg.com/vi/q3HtUfBLtQI/maxresdefault.jpg>

"1280"@en

"720"@en

"This presentation was recorded at GOTO EDA Day Warsaw 2024. #GOTOcon
#GOTOedahttps://gotopia.techEmily Shea - Head of Application Integration Go-To-Market
at..."@en

"544007664"@en

"YouTube"@en

"vnd.youtube://www.youtube.com/watch?v=q3HtUfBLtQI&feature=applinks"@en

"vnd.youtube://www.youtube.com/watch?v=q3HtUfBLtQI&feature=applinks"@en

"<http://www.youtube.com/watch?v=q3HtUfBLtQI&feature=applinks>"@en

"video.other"@en

"<https://www.youtube.com/embed/q3HtUfBLtQI>"@en

"<https://www.youtube.com/embed/q3HtUfBLtQI>"@en

"text/html"@en

"1280"@en

"720"@en

"YouTube"@en

"com.google.android.youtube"@en

"GOTO"@en

"GOTOcon"@en

Observație:
complementar, faptele (meta-datele) privind
resursele Web pot fi exprimate via JSON-LD



AgentCraft Hackathon Track 🎓

In 10 days it is happening!



DIAMANTAI
NOV 04, 2024



1



Hey Dear Community,

In just 10 days, it's happening! 🚀 On
conjunction with LangChain, is kicking off
only 7 days left before registration closes

JSON-LD

RDFa

RSS

POSH

Fact Collection #1

Subject

[#NewsArticle](#)

Predicates

[rdf:type](#)

[schema:NewsArticle](#)

[schema:author](#)

[#Person](#)

[schema:dateModified](#)

2024-11-04T15:02:54+02:00([schema>Date](#))

[schema:datePublished](#)

2024-11-04T15:02:54+02:00([schema>Date](#))

[schema:description](#)

In 10 days it is happening!

[schema:headline](#)

AgentCraft Hackathon Kickoff + Educational Track 🎓

[schema:image](#)

[#ImageObject_1](#)

[schema:isAccessibleForFree](#)

true([xsd:boolean](#))

[schema:mainEntityOfPage](#)

<https://diantantai.substack.com/p/agentcraft-hackathon-kickoff-educational>

[schema:publisher](#)

[#Organization](#)

[schema:url](#)

<https://diantantai.substack.com/p/agentcraft-hackathon-kickoff-educational>

Fact Collection #2

Subject

[#Person](#)

Predicates

[rdf:type](#)

[schema:Person](#)

[schema:description](#)

Creating the world's biggest practical GenAI knowledge hub

[schema:identifier](#)

user:186750220

[schema:image](#)

[#ImageObject](#)

[schema:name](#)

DiamantAI

[schema:url](#)

<https://substack.com/@diantantai>

Fact Collection #3

Subject

[#ImageObject](#)

Predicates

[rdf:type](#)

[schema:ImageObject](#)

fapte referitoare la un articol de știri (recurgerea
la conceptele **NewsArticle Person ImageObject Date**
specificate de schema.org) definite via JSON-LD



[schema:name](#)
[schema:offers](#)

Casa individuala 4 camere Manastirea Vladiceni - Tomesti
[#Offer](#)

Fact Collection #8

Subject [#PostalAddress](#)

Predicates

rdf:type	schema:PostalAddress
schema:addressCountry	Romania
schema:addressLocality	Iaşi
schema:addressRegion	Iaşi

Fact Collection #9

Subject [#Offer](#)

Predicates

rdf:type	schema:Offer
schema:availability	https://schema.org/InStock
schema:itemCondition	https://schema.org/NewCondition
schema:offeredBy	#RealEstateAgent
schema:price	167000
schema:priceCurrency	EUR
schema:url	https://www.imobiliare.ro/vanzare-individuale/iasi/tomesti/casa-individuala-de-vanzare-4-camere-XDRU11015?lista=327429439

Fact Collection #10

Subject [#RealEstateAgent](#)

Predicates

rdf:type	schema:RealEstateAgent
--------------------------	--

fapte vizând
domeniul imobiliar:
RealEstateAgent
Offer PostalAddress

 JSON-LD Input

 Options

```
{
  "@type": "Product",
  "name": "Casa individuala 4 camere Manastirea Vladiceni – Tomesti",
  "@context": "http://schema.org",
  "address": {
    "addressRegion": "Ia#351;i",
    "addressLocality": "Ia#351;i",
    "addressCountry": "Romania",
    "@type": "PostalAddress"
  },
  "description": "Cas#259; de v#226;rc#224;zare... an constr. 2023 (finalizata)",
  "image": "https://img3.imonet.ro/.../7966_620x465.jpg",
  "offers": {
    "@type": "Offer",
    "priceCurrency": "EUR",
    "price": "167000",
    "url": "https://www.imobiliare.ro/.../?lista=327429439",
    "availability": "https://schema.org/InStock",
    "itemCondition": "https://schema.org/NewCondition",
    "offeredBy": {
      "@type": "RealEstateAgent",
      "name": "REAL ESTATE",
      "url": "https://www.imobiliare.ro/agentii/..."
    }
  }
}
```

datele JSON-LD incluse în documentul HTML via
<script type="application/ld+json">...</script>

Expanded
Compacted
Flattened
Framed
N-Quads
Canonized
Table
Visualized
Signatures

```

_:b0 <http://schema.org/address> _:b1 .
_:b0 <http://schema.org/description> "Cas&#259; de v&acirc;nzare... an constr. 2023 (finalizata)" .
_:b0 <http://schema.org/image> <https://img3.imonet.ro/.../7966_620x465.jpg> .
_:b0 <http://schema.org/name> "Casa individuala 4 camere Manastirea Vladiceni – Tomesti" .
_:b0 <http://schema.org/offers> _:b2 .
_:b0 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://schema.org/Product> .
_:b1 <http://schema.org/addressCountry> "Romania" .
_:b1 <http://schema.org/addressLocality> "Ia&#351;i" .
_:b1 <http://schema.org/addressRegion> "Ia&#351;i" .
_:b1 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://schema.org/PostalAddress> .
_:b2 <http://schema.org/availability> "https://schema.org/InStock" .
_:b2 <http://schema.org/itemCondition> "https://schema.org/NewCondition" .
_:b2 <http://schema.org/offeredBy> _:b3 .
_:b2 <http://schema.org/price> "167000" .
_:b2 <http://schema.org/priceCurrency> "EUR" .
_:b2 <http://schema.org/url> <https://www.imobiliare.ro/.../?lista=327429439> .
_:b2 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://schema.org/Offer> .
_:b3 <http://schema.org/name> "REAL ESTATE" .
_:b3 <http://schema.org/url> <https://www.imobiliare.ro/agentii/...> .
_:b3 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://schema.org/RealEstateAgent> .

```

aceleași date convertite în formatul RDF *N-quads*
de către aplicația Web **JSON-LD Playground**
json-ld.org/playground/

Statement Collection #1

Subject

#Movie

Predicates

rdf:typeschema:Movieschema:actor

#Person

schema:actor

#Person_5

schema:actor

#Person_6

schema:aggregateRating

#AggregateRating

schema:contentRating

R

schema:creator

#Person_1

schema:creator

#Organization

schema:creator

#Organization_1

schema:creator

#Organization_2

schema:creator

#Person_7

schema:creator

#Person_8

schema:datePublished1992-04-03(schema>Date)schema:description

Post-apocalyptic surrealist black comedy about the landlord occasionally prepares a delicacy for his odd tenants.

schema:director

#Person_2

schema:director

#Person_3

schema:duration

PT1H39M

schema:genre

Comedy

schema:genre

Crime

meta-date despre o producție cinematografică:
se recurge la conceptul **Movie** oferit de modelul schema.org

Statement Collection #4

Subject [#Person_2](#)

Predicates

[rdf:type](#) [schema:Person](#)

[schema:name](#) Marc Caro

[schema:url](#) <https://www.imdb.com/name/nm0001988/>

Statement Collection #5

Subject [#Person_3](#)

Predicates

[rdf:type](#) [schema:Person](#)

[schema:name](#) Jean-Pierre Jeunet

[schema:url](#) <https://www.imdb.com/name/nm0000466/>

Statement Collection #6

Subject [#Review](#)

Predicates

[rdf:type](#) [schema:Review](#)

[schema:author](#) [#Person_4](#)

[schema:dateCreated](#) 2004-09-09([schema:Date](#))

[schema:inLanguage](#) English

[schema:itemReviewed](#) [#CreativeWork](#)

[schema:name](#) One most inventive and original films to grace the silver screen

regizorii (obiecte precizate cu proprietatea **director**)
sunt instanțe de tip **Person**

USE

Căutare

JSON-LD

RDFa

POSH

CARTURESTI.RO / CARTE STRAINA /

Fact Collection #1

Subject

#Product

Predicates

rdf:type	schema:Product
schema:GTIN13	9780719842771
schema:brand	#Thing
schema:description	This beautiful book shows the artist how to develop their own unique way of seeing and interpreting light and colour in oils, using plein air painting as the foundation (painting outdoors direct from life). It serves as a catalyst for exploration, both out in the field, and back in the studio. Packed with finished examples and step-by-step sequences, it guides you through the beguiling challenge of painting the light, demonstrating how to say 'just enough' in your own work.
schema:image	https://cdn.dc5.ro/img-prod/2368415140-2-240.jpeg
schema:name	Painting Light and Colour in Oils
schema:offers	#Offer
schema:sku	9780719842771

Fact Collection #2

Subject

[#Thing](#)

Predicates

rdf:type	schema:Thing
schema:name	The Crowood Press Ltd

Fact Collection #3

Subject

[#Offer](#)

Predicates

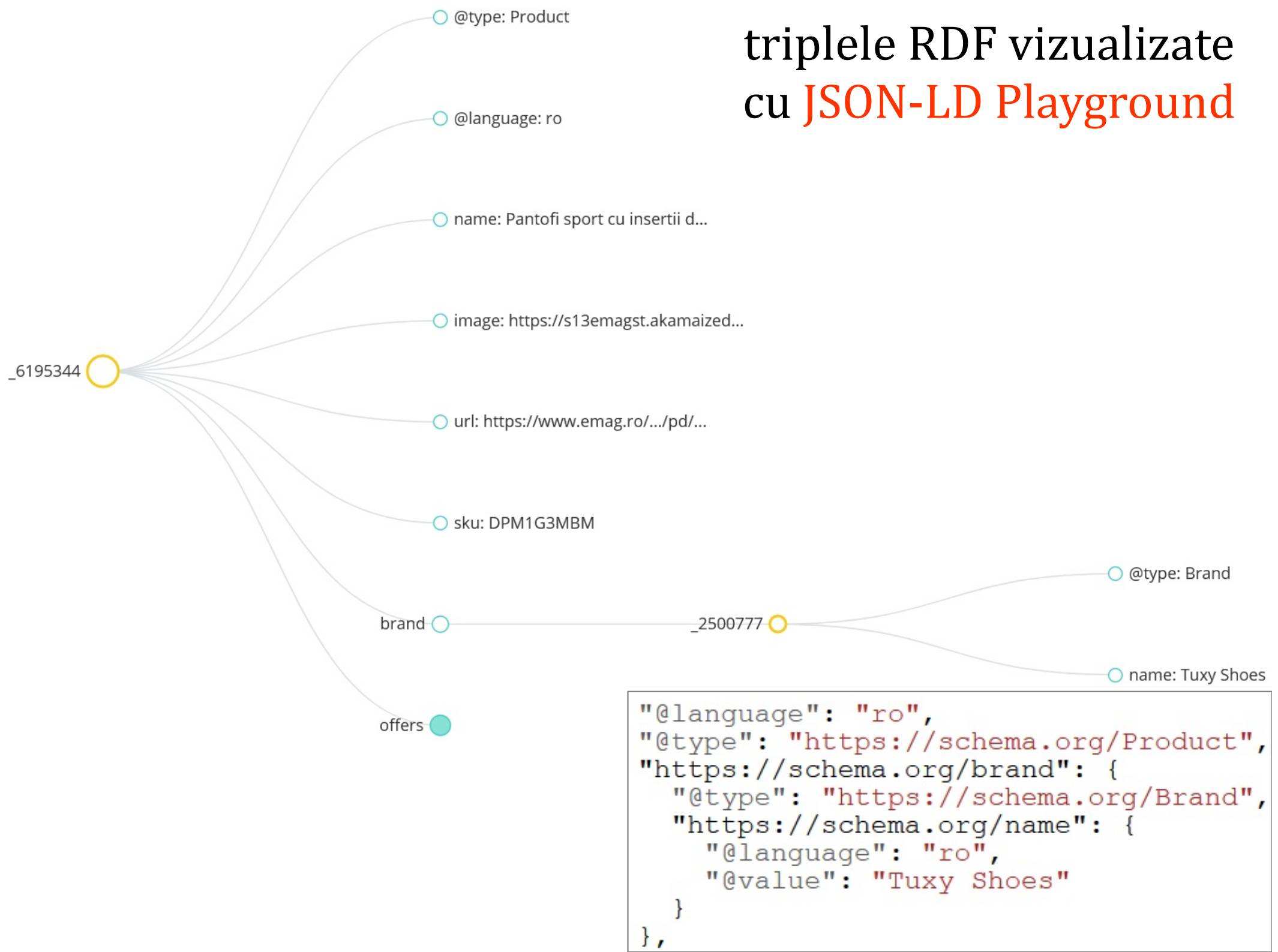
rdf:type	schema:Offer
schema:availability	http://schema.org/OutOfStock
schema:itemCondition	http://schema.org/NewCondition
schema:price	145.96

meta-date despre o carte de artă – aici, o instanță a conceptului **Product** definit de schema.org –, recurgându-se la proprietățile **brand description image name** etc. și cele vizând o ofertă (**Offer + availability itemCondition price**)

```
{
  "@context": { "@vocab": "https://schema.org/", "@language": "ro" },
  "@type": "Product",
  "description": "...",
  "name": "Pantofi sport cu inserții de RDF 😊",
  "image": "https://s1emagst.akamaized.net/.../666ce3.jpg",
  "url": "https://www.emag.ro/.../pd/DPM1G3MBM/",
  "sku": "DPM1G3MBM",
  "brand": {
    "@type": "Brand",
    "name": "Tuxy Shoes"
  },
  "offers": {
    "@type": "Offer",
    "seller": "eMAG",
    "availability": "http://schema.org/InStock",
    "price": "349.99",
    "priceCurrency": "RON"
  }
}
```

meta-date despre un produs
oferit de emag.ro
extrase și serializate
ca obiect JSON-LD
(adaptare după codul HTML real)

triplele RDF vizualizate cu JSON-LD Playground



```
{
  "@context": "https://schema.org",
  "@type": "BreadcrumbList",
  "itemListElement": [
    {
      "@type": "ListItem",
      "position": 1,
      "item": {
        "@id": "https://www.emag.ro/fashion/d",
        "name": "Fashion"
      }
    },
    {
      "@type": "ListItem",
      "position": 2,
      "item": {
        "@id": "https://www.emag.ro/incaltaminte-barbati/sd",
        "name": "Incaltaminte barbati"
      }
    }
  ]
}
```

exprimarea în JSON-LD a metodelor de
navigabilitate Web via schema.org

Toate categoriile

Discuri vir

JSON-LD

RDFa

POSH

/ Discuri LP și CD-uri / CD-uri muzica

Classical - CD

1 - 34 de 283

Toate filtrele

În stoc ^

☐ Disponibile imediat (213)

Preț ^

Fact Collection #1

Subject#BreadcrumbList

Predicates

rdf:type
schema:itemListElement
schema:itemListElement

schema:BreadcrumbList
#ListItem
#ListItem_1

Fact Collection #2

Subject#ListItem

Predicates

rdf:type
schema:item
schema:name
schema:position

schema:ListItem
<https://www.muziker.ro/discuri-vinil-si-cd-uri>
Discuri LP și CD-uri
1([xsd:integer](#))

Fact Collection #3

Subject#ListItem_1

Predicates

rdf:type
schema:item
schema:name
schema:position

schema:ListItem
<https://www.muziker.ro/cd-uri-muzica>
CD-uri muzica
2([xsd:integer](#))

Discuri vir

JSON-LD

RDFa

POSH

/ Discuri LP și CD-uri / CD-uri muzica

Classical - CD

1 - 34 de 283

 Toate filtrele

În stoc

☐ Disponibile immediat (213)

Preț $\wedge$

Fact Collection #1

Subject

#BreadcrumbList

Predicates

rdf:type

schema:BreadcrumbList

[schema:itemListElement](#)

#ListItem

[schema:itemListElement](#)

#ListItem_1

Fact Collection #2

Subject

#ListItem

Predicates

rdf:type

schema:ListItem

[schema:item](#)

<https://www.muziker.ro/discuri-vinil-si-cd-uri>

schema:name

Discuri LP și CD-uri

schema:position

1(xsd:integer)

Fact Collection #3

Subject

#ListItem_1

Predicates

rdf:type

schema:ListItem

schema:item

<https://www.muziker.ro/cd-uri-muzica>

schema:name

CD-uri muzica

schema:position

2(xsd:integer)

alt exemplu concret

Why Apple Watch

JSON-LD

RDFa

POSH

Predicates

rdf:type	schema:ContactPoint
schema:areaServed	US
schema:availableLanguage	EN
schema:availableLanguage	ES
schema:contactType	customer support
schema:telephone	+1-800-275-2273

Fact Collection #6

Subject

[#product](#)

Predicates

rdf:type	schema:Product
schema:brand	https://www.apple.com/watch/#brand
schema:description	When you wear Apple Watch, you get a fitness partner, meaningful health insights, safety features, and a connection to what you care about most.
schema:image	

Fact Collection #7

Subject

[#BuyAction](#)

Predicates

rdf:type	schema:BuyAction
schema:seller	https://www.apple.com/#organization
schema:url	https://www.apple.com/us/shop/goto/buy_watch

[schema:mainEntityOfPage](#)

<https://www.apple.com/watch/>

[schema:manufacturer](#)

<https://www.apple.com/#organization>

[schema:name](#)

Watch

[schema:potentialAction](#)

[#BuyAction](#)

[schema:url](#)

<https://www.apple.com/watch/>

specificarea cu JSON-LD a declarațiilor RDF vizând instanțe de concepte **ContactPoint** (**areaServed availableLanguage**) și **Product** (**brand mainEntityOfPage manufacturer**) și o posibilă acțiune: **BuyAction** cu proprietățile **seller url**

WHAT HI-FI? THE WORLD'S #1 TECH BUYER'S GUIDE

[Home](#)
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[Best Buys](#)
[Reviews](#)
[Hi-Fi](#)
[Headphones](#)

TRENDING
[What Hi-Fi? Awards 2025](#)
[Arcam](#)

[Speakers](#) > [Wireless Speakers](#)

Best AirPlay speakers 2025: wireless streaming for Apple devices tried and tested

Best Buys
By [Harry McKerrell](#) last updated 1 hour ago

The best AirPlay speakers tested by our experts

When you purchase through links on our site, we may earn an affiliate commission.

Microdata
JSON-LD
RDFa
RSS
POSH

Fact Collection #4 Subject Predicates rdf:type schema:xpath schema:xpath	#SpeakableSpecification schema:SpeakableSpecification /html/head/meta[@name="description"]/@content /html/head/title
Fact Collection #5 Subject Predicates rdf:type schema:mainEntity schema:mainEntity schema:mainEntity schema:name	#FAQPage schema:FAQPage #Question #Question_1 #Question_2 Best AirPlay speakers 2025: wireless streaming for Apple devices tried and tested
Fact Collection #6 Subject Predicates rdf:type schema:acceptedAnswer schema:name	#Question schema:Question #Answer Is AirPlay better than Bluetooth?
Fact Collection #7 Subject Predicates rdf:type	#Answer schema:Answer

precizarea în JSON-LD a manierei de interacțiune vocală cu un sit Web via conceptul **SpeakableSpecification** + proprietatea **xpath** (aici, se utilizează expresii XPath pentru a selecta conținuturi textuale de interes)

Fact Collection #1

Subject	#Hotel
Predicates	
rdf:type	schema:Hotel
schema:address	#PostalAddress
schema:checkinTime	3:00 PM
schema:checkoutTime	12:00 PM
schema:description	A perfect choice for business or leisure, our Hampton by Hilton Iasi city centre hotel offers modern guest rooms within minutes from all major attractions.
schema:geo	#GeoCoordinates
schema:image	
schema:name	Hampton by Hilton Iasi
schema:openingHours	Mo, Tu, We, Th, Fr, Sa, Su
schema:telephone	+40 232 242 000
schema:url	https://www.hilton.com/hotels/iashxhx-hampton-iasi/

Fact Collection #2

Subject	#PostalAddress
Predicates	
rdf:type	schema:PostalAddress
schema:addressCountry	RO
schema:addressLocality	Iasi
schema:postalCode	700020
schema:streetAddress	St. Anastasie Panu 26, Iasi, 700020, Romania

(meta-)date JSON-LD
referitoare la un hotel



Statement Collection #1

Subject <https://www.drugs.com/#organization>

Predicates

[rdf:type](#) [schema:Organization](#)

[schema:logo](#) [#ImageObject_1](#)

[schema:name](#) Drugs.com

[schema:url](#) <https://www.drugs.com>

Statement Collection #2

Subject https://www.drugs.com/drug_interactions.html

Predicates

[rdf:type](#) [schema:MedicalWebPage](#)

[schema:audience](#) [#medicalAudience](#)

[schema:description](#) Check for multi-drug interactions including alcohol, food, supplements & diseases. Includes detailed reports for both patients and health professionals.

[schema:image](#) [#ImageObject](#)

[schema:name](#) Drug Interactions Checker

[schema:publisher](#) <https://www.drugs.com/#organization>

[schema:url](#) https://www.drugs.com/drug_interactions.html

Statement Collection #3

Subject [#medicalAudience](#)

Predicates

[rdf:type](#) [schema:medicalAudience](#)

[schema:audienceType](#) Patient

declarații JSON-LD despre pagini Web cu conținut medical
și persoanele vizate (**MedicalWebPage medicalAudience**)

Memeland
Moonit
Potatoz

JSON-LD
RDFa
POSH


Humor
support • 3h

French Royalty in the 18th century



Fact Collection #1

Subject

Predicates

[rdf:type](#)

[schema:author](#)

[schema:datePublished](#)

[schema:description](#)

[schema:headline](#)

[schema:image](#)

<https://9gag.com/gag/aqyB7RR>

[schema:SocialMediaPosting](#)

[#Person](#)

2025-10-27T07:35:06+00:00([schema:Date](#))

This meme features two cats, one peeking out of a basket and the other with its rear facing the camera, accompanied by the text "French Royalty in the 18th century:". The meme uses humor to comment on the perceived extravagance and detachment of French royalty, particularly referencing the famous (though likely apocryphal) quote "Let them eat cake" attributed to Marie Antoinette. The cat peeking out of the basket represents the oblivious or detached royalty, while the cat with its rear facing the camera, seemingly ignoring the other, adds to the comedic effect of indifference or self-absorption.

- 9GAG

French Royalty in the 18th century:



construcții JSON-LD vizând un articol oferit de o aplicație Web socială (conceptul **SocialMediaPosting** și proprietățile asociate: **author datePublished description headline image**)

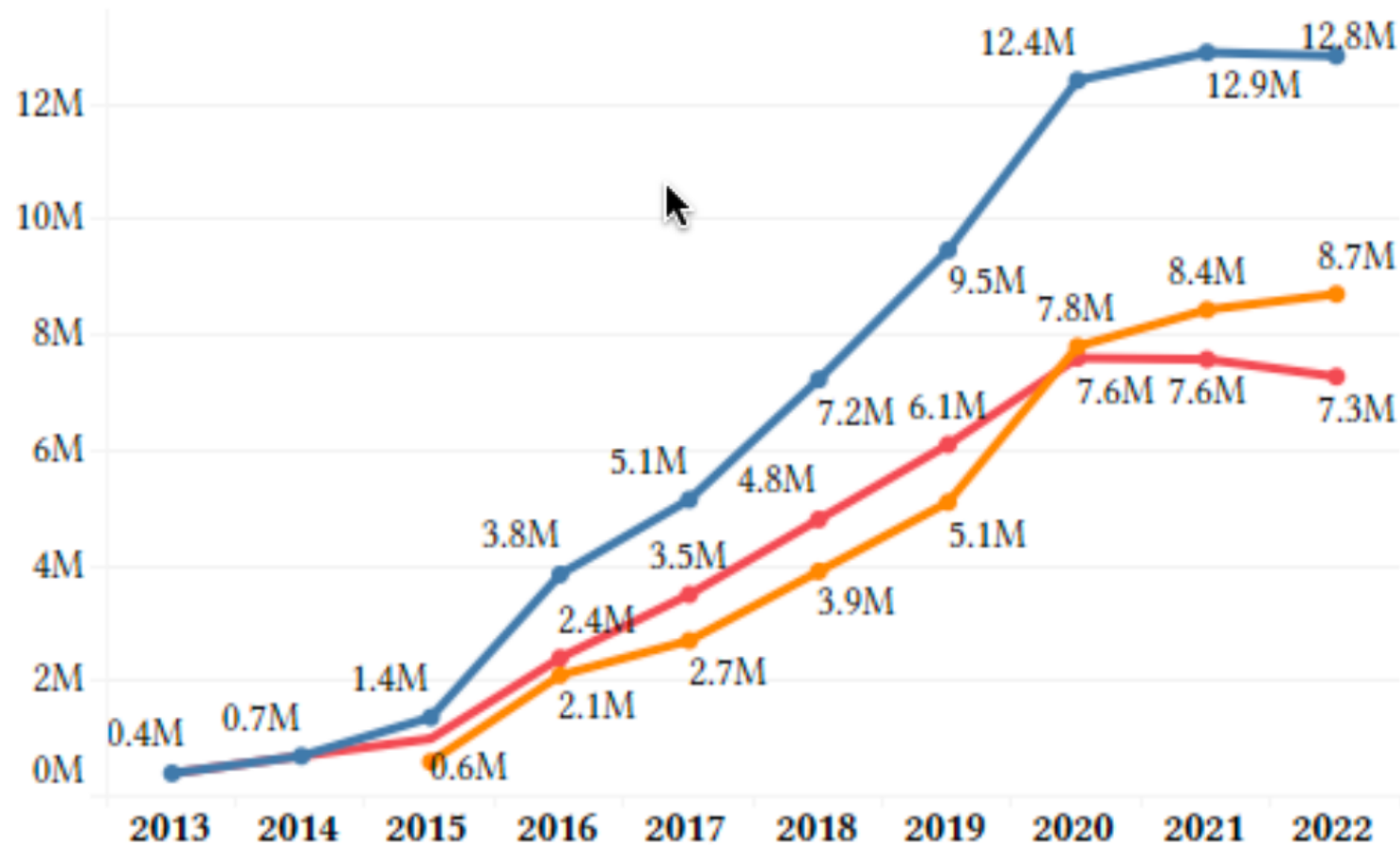
Aserțiunile RDFa sunt indexate de motoarele de căutare

Bing Webmaster

www.bing.com/webmasters/help/marking-up-your-site-with-structured-data-3a93e731

Google Structured Data

developers.google.com/search/docs/appearance/structured-data/



Web Data Commons

webdatacommons.org/structureddata/

aici: studiu asupra evoluției adnotărilor semantice (2013–2022)

A. Brinkmann, A. Primpeli, C. Bizer, “The Web Data Commons Schema.org Data Set Series”, *WWW’23 Companion*, ACM (2023)

microformat → microdate → RDFa

Microformatele pot fi exprimate
via microdate HTML care, la rândul lor,
pot avea asociate construcții RDF(a)

procesul de conversie microdate HTML → RDF
este detaliat la www.w3.org/TR/microdata-rdf/

microformat → microdata → RDFa

Cel mai expresiv – general – model este RDF(a)

```

<li class="conference" itemscope="" itemid="https://wade.info/2021/"
    itemtype="http://schema.org/SocialEvent
        http://schema.org/BusinessEvent http://schema.org/EducationEvent">
    <h2><a itemprop="url" href="https://wade.info">
        <span itemprop="name">WADe 2021 Fictional Event</span></a></h2>
    <p itemprop="location" itemscope=""
        itemid="https://wade.info/venues/London"
        itemtype="http://schema.org/Place">
        <span itemprop="name">
            <a href="https://www.wikidata.org/wiki/Q145">UK</a> in
            <a itemprop="url"
                href="https://www.wikidata.org/wiki/Q84">London</a></span></p>
    <p class="date">
        <time itemprop="startDate" datetime="2021-10-25">25 Oct.</time> –
        <time itemprop="endDate" datetime="2021-11-05">5 Nov.</time>
    </p>
    ...
</li>

```

modelarea unui eveniment pe baza schema.org
 + resurse specificate de Wikidata
 adaptare după Jeni Tennison

@prefix s: <<http://schema.org>>

<<https://wade.info/2021/>> a s:SocialEvent , s:BusinessEvent ,
s:EducationEvent ;
s:url <<https://wade.info>> ;
s:name "WADe 2021 Fictional Event" ;
s:location <<https://wade.info/venues/London>> ;
s:startDate "2021-10-25"^^xsd:date ;
s:endDate "2021-11-05"^^xsd:date ;

precizare explicită
a tipului de date

<<https://wade.info/venues/London>> a s:Place ;
s:url <<https://www.wikidata.org/wiki/Q84>> ;
s:name "London (UK)" .

referire la resurse
(entități) Wikidata

construcțiile RDF (în format Turtle) echivalente
pe baza asocierilor (*mappings*) realizate

@id	http://schema.org/BusinessEvent(1)										
rdf:type	rdfs:Class										
@id	http://schema.org/EducationEvent(1)										
rdf:type	rdfs:Class										
@id	http://schema.org/Place(1)										
rdf:type	rdfs:Class										
@id	http://schema.org/SocialEvent(1)										
rdf:type	rdfs:Class										
@id	https://wade.info/2021/(0)										
rdf:type	rdfs:Resource schema:Thing schema:Event schema:BusinessEvent schema:SocialEvent schema:EducationEvent										
rdfs:label	WADe 2021 Fictional Event										
schema:endDate	Friday, 05 November 2021										
schema:location	<table> <tr> <td>@id</td><td>https://wade.info/venues/London(1)</td></tr> <tr> <td>rdf:type</td><td>rdfs:Resource schema:Thing schema:Place</td></tr> <tr> <td>rdfs:label</td><td>UK in London</td></tr> <tr> <td>schema:name</td><td>UK in London</td></tr> <tr> <td>schema:url</td><td>https://www.wikidata.org/wiki/Q84</td></tr> </table>	@id	https://wade.info/venues/London(1)	rdf:type	rdfs:Resource schema:Thing schema:Place	rdfs:label	UK in London	schema:name	UK in London	schema:url	https://www.wikidata.org/wiki/Q84
@id	https://wade.info/venues/London(1)										
rdf:type	rdfs:Resource schema:Thing schema:Place										
rdfs:label	UK in London										
schema:name	UK in London										
schema:url	https://www.wikidata.org/wiki/Q84										
schema:name	WADe 2021 Fictional Event										
schema:startDate	Monday, 25 October 2021										
schema:url	https://wade.info										

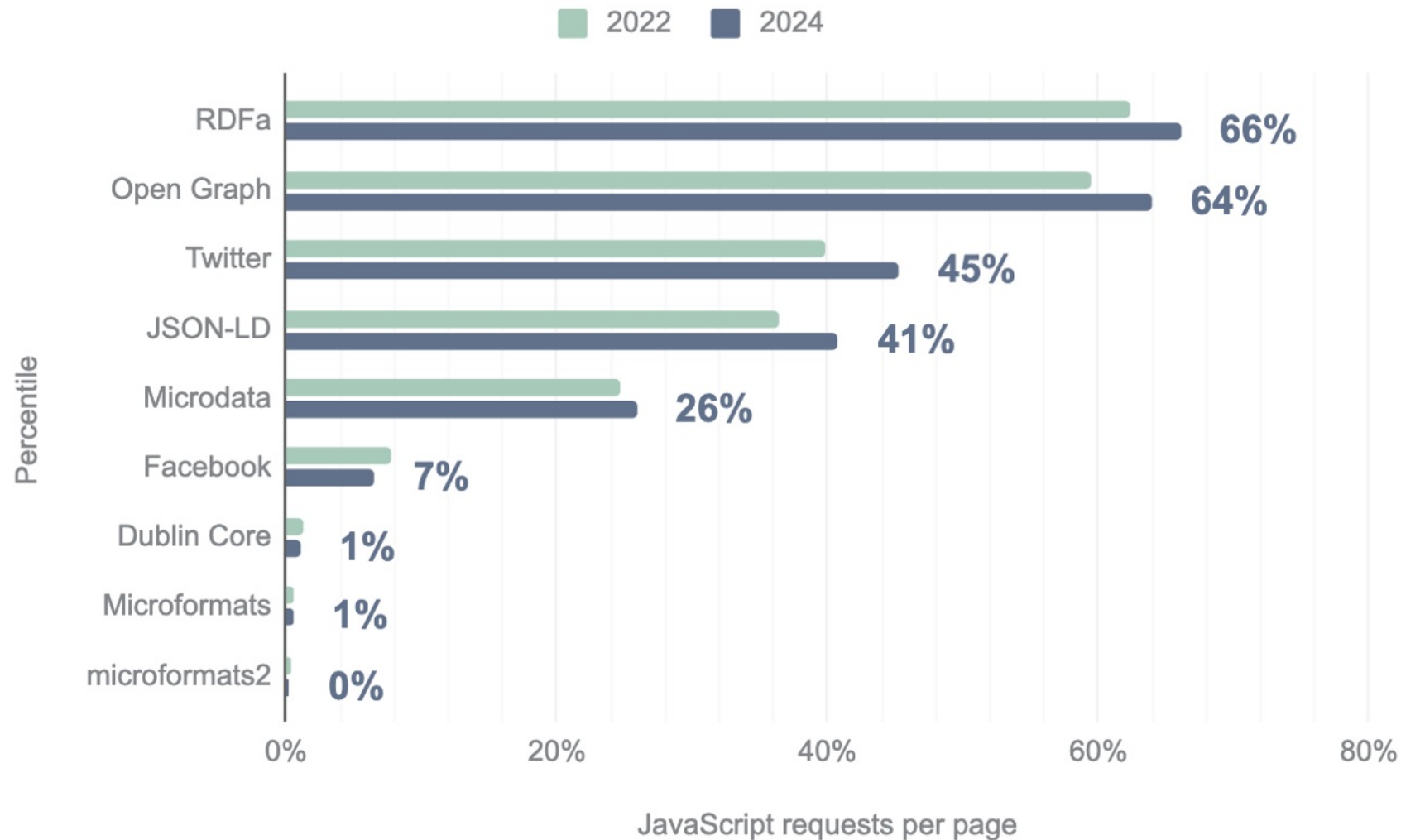
datele RDFa verificate și extrase via instrumentul Web
Structured Data Linter (implementat în Ruby)

Bibliotece de procesare JSON-LD
(C#, Go, JavaScript, Java, Python, PHP, Rust, TypeScript):

json-ld.org/#developers

Structured data usage by year

Web Almanac 2024: Structured Data (mobile)

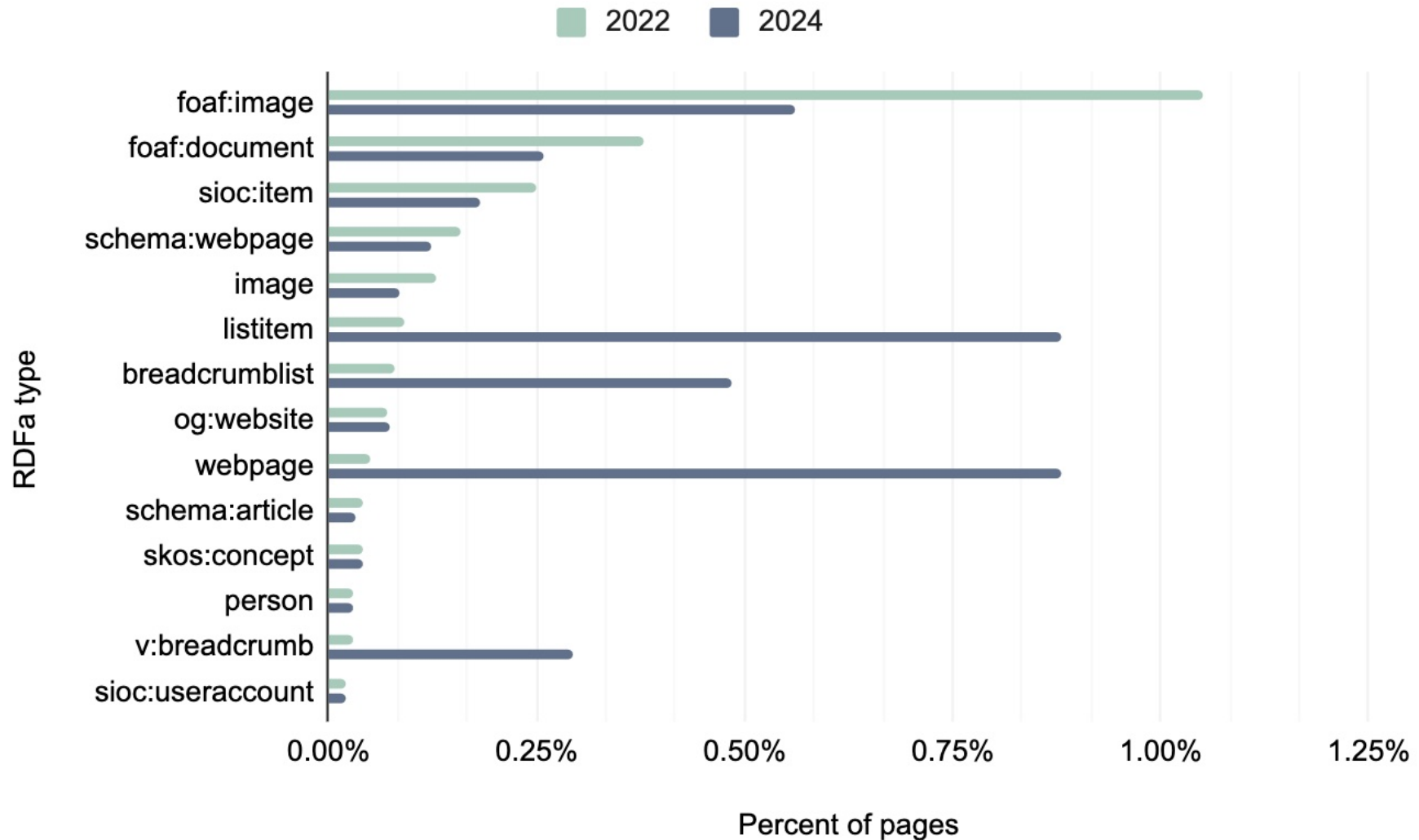


recurgerea la date structurate în contextul dispozitivelor mobile – de citit Andrea Volpini, „Structured Data”,
Web Almanac 2024

almanac.httparchive.org/en/2024/structured-data

RDFa usage by year - Desktop

Web Almanac 2024: Structured Data

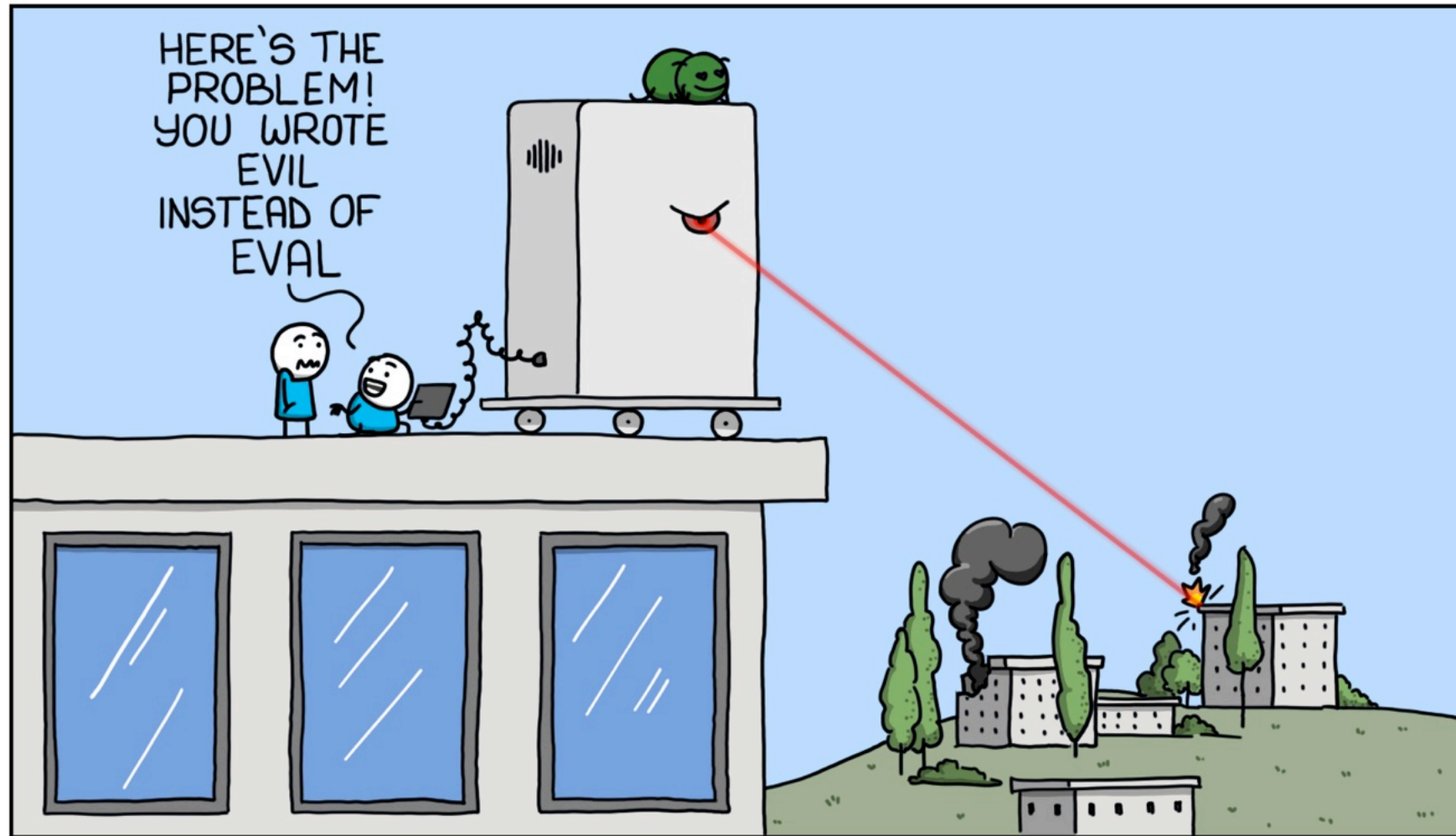


utilizarea construcțiilor RDFa la nivel de *desktop* – detalii:
almanac.httparchive.org/en/2024/structured-data

rezumat

NATURAL LANGUAGE INSTRUCTIONS

MONKEYUSER.COM



RDFA – specificarea triplelor RDF direct în HTML